

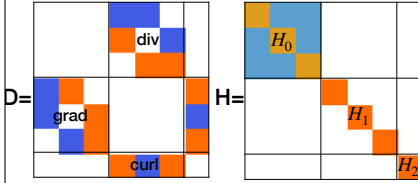
Euler-Poincaré

G: finite abstract simplicial complex.

$\Lambda = l^2(G)$ space of **differential forms**. Linear space of dim = |G|.

If $y \subset x$, $\dim(y)=\dim(x)-1$, define $d(x,y) = \text{sign}(x,y)$. Else $d(x,y)=0$. d is called the **exterior derivative**.

$D = d + d^*$ Dirac operator
 $H = D^2 = dd^* + d^*d$ Hodge
 $H = \bigoplus_{k=0}^q H_k$ block decomposition
 $b_k = \dim(\ker(H_k))$ Betti numbers



Euler-Poincaré



$$\chi(G) = \sum_{k=0}^q (-1)^k b_k$$

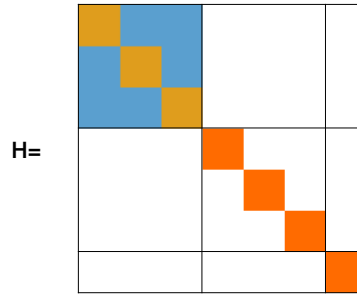


McKean-Singer

$$\text{str}(H) = \sum_{k=0}^q (-1)^k \text{tr}(H_k) \text{ super trace}$$

$$\text{str}(e^{-tH}) = \chi(G)$$

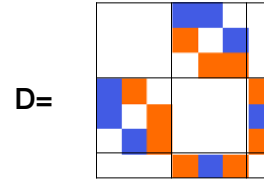
$$G = \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$



$$\begin{aligned} \sigma(H_0) &= \{0, 3, 3\} & H_0 &= \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \\ \sigma(H_1) &= \{3, 3, 3\} & H_1 &= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \\ \sigma(H_2) &= \{3\} & H_2 &= [3] \end{aligned}$$

Proof of MacKean-Singer

$$\begin{aligned} \Lambda &= \ker(H) \oplus \text{im}(H) \\ \text{im}(H) &= \text{im}(d) \oplus \text{im}(d^*) \\ \Lambda &= \Lambda_{\text{even}} \oplus \Lambda_{\text{odd}} \end{aligned}$$



$D : \Lambda_{\text{even}} \rightarrow \Lambda_{\text{odd}}$
 defines an isomorphism from $\text{im}(H)_{\text{even}} \rightarrow \text{im}(H)_{\text{odd}}$

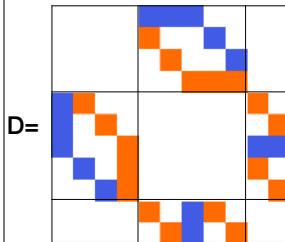
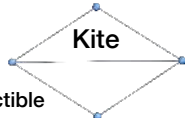
$$\begin{aligned} \text{str}(L^n) &= 0, \forall n > 0 \\ \text{str}(e^{-tH}) &= \text{str}(1 + tH + \frac{H^2}{2}t^2 + \dots) \\ &= \text{str}(1) = \chi(G) \end{aligned} \quad \text{QED}$$

$$G = \{\{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,4\}, \{3,4\}, \{1,2,4\}, \{1,3,4\}\}$$

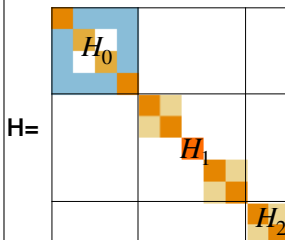
$$G = K_{1,2,1}$$

$$b=(1,0,0)$$

(b_0, b_1, b_2) contractible



$$\begin{aligned} \sigma(D) &= \\ &= \{-2, -2, -2, 2, 2, 0, \\ &= -\sqrt{2}, -\sqrt{2}, \sqrt{2}, \sqrt{2}\} \end{aligned}$$



$$\sigma(H_0) = \{4, 4, 2, 0\}$$

$$\sigma(H_1) = \{4, 4, 4, 2, 2\}$$

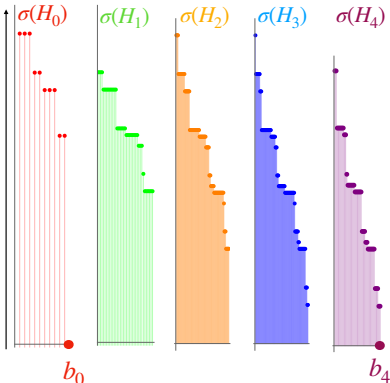
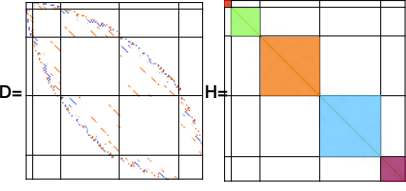
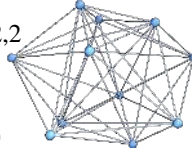
$$\sigma(H_2) = \{4, 2\}$$

$$G = K_5 \oplus K_{2,2,2}$$

4-sphere

$$b=(1,0,0,0,1)$$

$$(b_0, b_1, b_2, b_3, b_4)$$



Proof of Euler Poincaré

McKean-Singer showed that $\text{str}(e^{-tH}) = \text{str}(1 + tH + \frac{H^2}{2}t^2 + \dots)$ is invariant of t. For t=0 it is

$$\text{str}(1) = \sum_{k=0}^q (-1)^k f_k = \chi(G)$$

$H = D^2$ has non-negative spectrum.

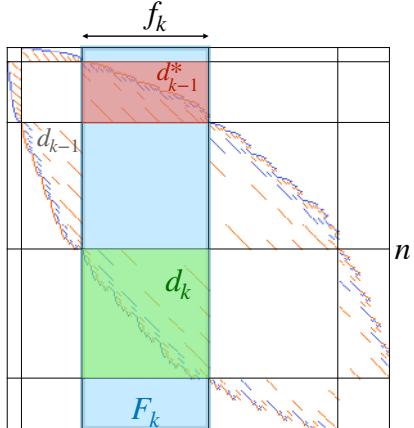
For every eigenvector f of H with positive eigenvalue λ , we have

$$e^{-tH}f = e^{-t\lambda}f \rightarrow 0$$

Therefore $P = \lim_{t \rightarrow \infty} e^{-tH}$ is the projection on $\ker(H)$. But

$$\text{str}(P) = \sum_{k=0}^q (-1)^k b_k \quad \text{QED}$$

Hodge picture



$$\begin{aligned} F_k^* F_k &= H_k & \ker(F_k) &= \ker(H_k) \\ &= \ker(d_k) \cap \ker(d_{k-1}^*) \\ \text{im}(d_{k-1}) &\subset \ker(d_k) \\ \ker(d_{k-1}^*)^\perp &\subset \ker(d_k) \\ \dim(\ker(d_k)/\text{im}(d_{k-1})) &= \dim(\ker(H_k)) \end{aligned}$$

Calabi-Yau example

G=K3 complex surface

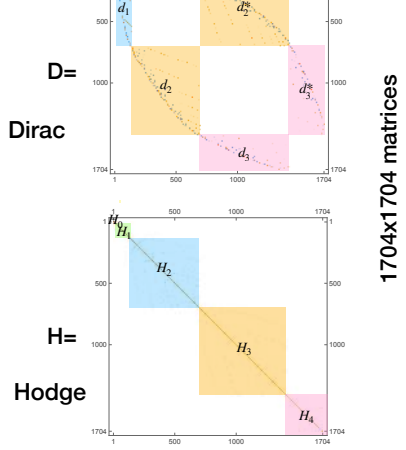
$$x^4 + y^4 + z^4 + w^4 = 0 \subset \mathbb{CP}^3$$

Calabi-Yau manifold of dimension 2

~ smooth real 4 manifold

$$f=(16, 120, 560, 720, 288)$$

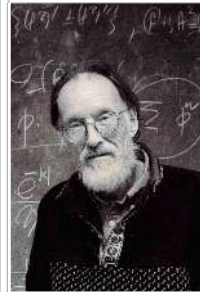
$$b=(1, 0, 22, 0, 1) \quad \chi(G) = 24$$



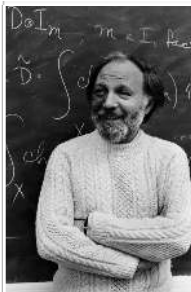
William Hodge
1903-1975



Paul Dirac
1902-1984



Henry McKean
1930-2024



Isadore Singer
1924-2021